

**ATOMIC-COLUMN COMPOSITION ANALYSIS OF GaAs/AlGaAs
SEMICONDUCTOR INTERFACES USING ANSCOMBE-DENOISED
HAADF-STEM IMAGES**

by

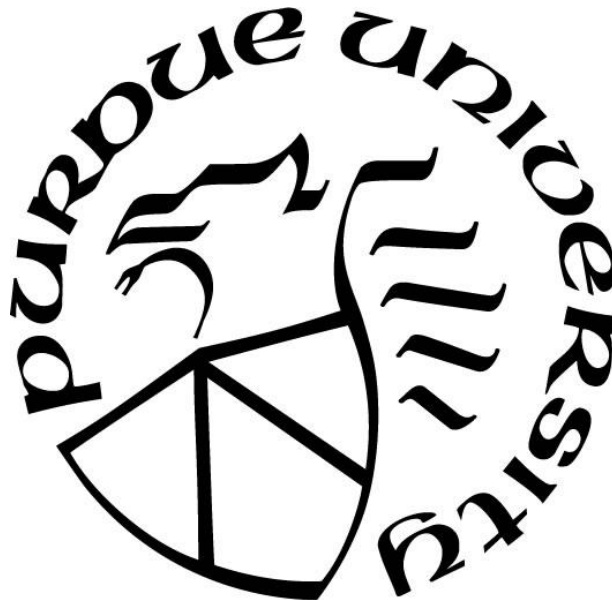
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Dedicated to my family, teachers, and friends

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ABSTRACT

Al/Ga-based semiconductor heterostructures, such as AlGaAs/GaAs, are widely used in electronic and optoelectronic devices. Their properties depend on interface roughness, Al and Ga composition, the thicknesses of quantum wells and barriers, carrier density, doping profile and impurity concentration, defects and dislocations, among other external factors. Changes in these key properties can affect band offsets, carrier confinement and carrier transport. The samples studied in this thesis are part of a collaboration with the Quantum Semiconductor Systems Group led by Professor Manfra. The analysis of the interfaces of these systems is important because small variations in the local Al/Ga composition can influence carrier scattering and device performance. Measuring the local composition at the atomic scale can therefore help relate interface chemistry to the behavior of the semiconductor heterostructure. In this thesis, a quantitative model is developed to extract local compositional information from high-angle annular dark-field (HAADF) scanning transmission electron microscopy (STEM) images of semiconductor heterostructures at the atomic scale. In HAADF-STEM, image intensity is directly related to the atomic number (Z) of the elements present in a sample, as well as sample thickness and imaging conditions. This gives rise to the so-called Z -contrast imaging mode. However, raw HAADF-STEM intensity is not easy to interpret as a direct measure of local composition at semiconductor interfaces. At these length scales, the measured contrast is altered by noise, probe broadening, finite probe size, and overlap with neighboring atomic columns. These effects prevent straightforward measurement of local Al/Ga composition from raw HAADF intensity alone. This motivates the use of quantitative image-analysis models to extract more reliable compositional information from the HAADF-STEM images. This thesis makes two main contributions. First, it compares preprocessing methods for HAADF-STEM data. The results show that Anscombe-based denoising is useful for this dataset because it reduces high-frequency noise while preserving lattice information needed for atomic-column fitting. Principal component analysis (PCA) based denoising is also evaluated but the results indicate that it can change the structural intensity information depending on the number of components used. Second, this thesis develops a model-based workflow to analyze atomic columns in HAADF-STEM images. The image is modeled as a superposition of atomic-column contributions convolved with the electron probe, such that a physically meaningful and column-specific peak volume can be fit in place of raw image brightness. These fitted quantities

are then used within a relative-composition framework to compare to known barrier and quantum well regions, and to estimate the composition of unknown interface columns. A second-pass correction is also applied to remove an empirical crosstalk contribution from adjacent cation columns into nearby As-column intensity. This correction is employed to produce an intensity-corrected ROI, to rerun the fitted-column workflow, and to generate an updated relative Al-fraction profile using a local cation/As intensity ratio. The corrected result is considered a refinement of the relative-composition analysis, not an independently calibrated absolute composition measurement. This work represents an improved method for atomic-column-level compositional mapping and a more robust approach for quantitative characterization of nanoscale semiconductor interfaces.

1. INTRODUCTION

1.1 Motivation

Semiconductor heterostructures are formed by depositing chemically different materials on top of each other. III-Vs such as GaAs/AlGaAs form quantum wells/barriers whose proper operation is determined not only by nominal layer design, but also by the quality of the interfaces on the atomic scale. A nominally abrupt interface may be smeared out by interdiffusion, local roughness, or chemical segregation. These effects alter the local composition through one or more atomic planes and affect device behavior on the electrical and optical level.

It is thus the main scientific driver of this thesis to go beyond visual inspection of a HAADF-STEM image and calculate a quantitative estimate of local interface composition. A direct interpretation by eye is not possible, because we do not measure individual interface columns in isolation. The observed brightness of each atomic column is changed by detector noise, probe broadening, overlap from neighboring columns, and possible thickness variation. Therefore, a professional quantitative workflow must detail how the image was preprocessed, how atomic columns were modeled, how fitted quantities were extracted from each column, and how those quantities were compared to known regions on the well or barrier sides of the interface.

This thesis follows a natural order of topics moving from concepts in electron optics to ultimately solve a computational problem. The initial work characterized what STEM contrast means physically: electrons are scattered by the specimen, high-angle scattering comes to the HAADF detector, and the intensity in an atomic column scale with the average atomic number. The next chapter treated the image as data: noise was quantified, structure was diagnosed using Fourier-domain line profiles, and Anscombe variance stabilization was contrasted with PCA-based denoising. Model-based fitting was used in the final chapter to extract peak volumes from each atomic column and then to compare unknown interface columns against well and barrier references.

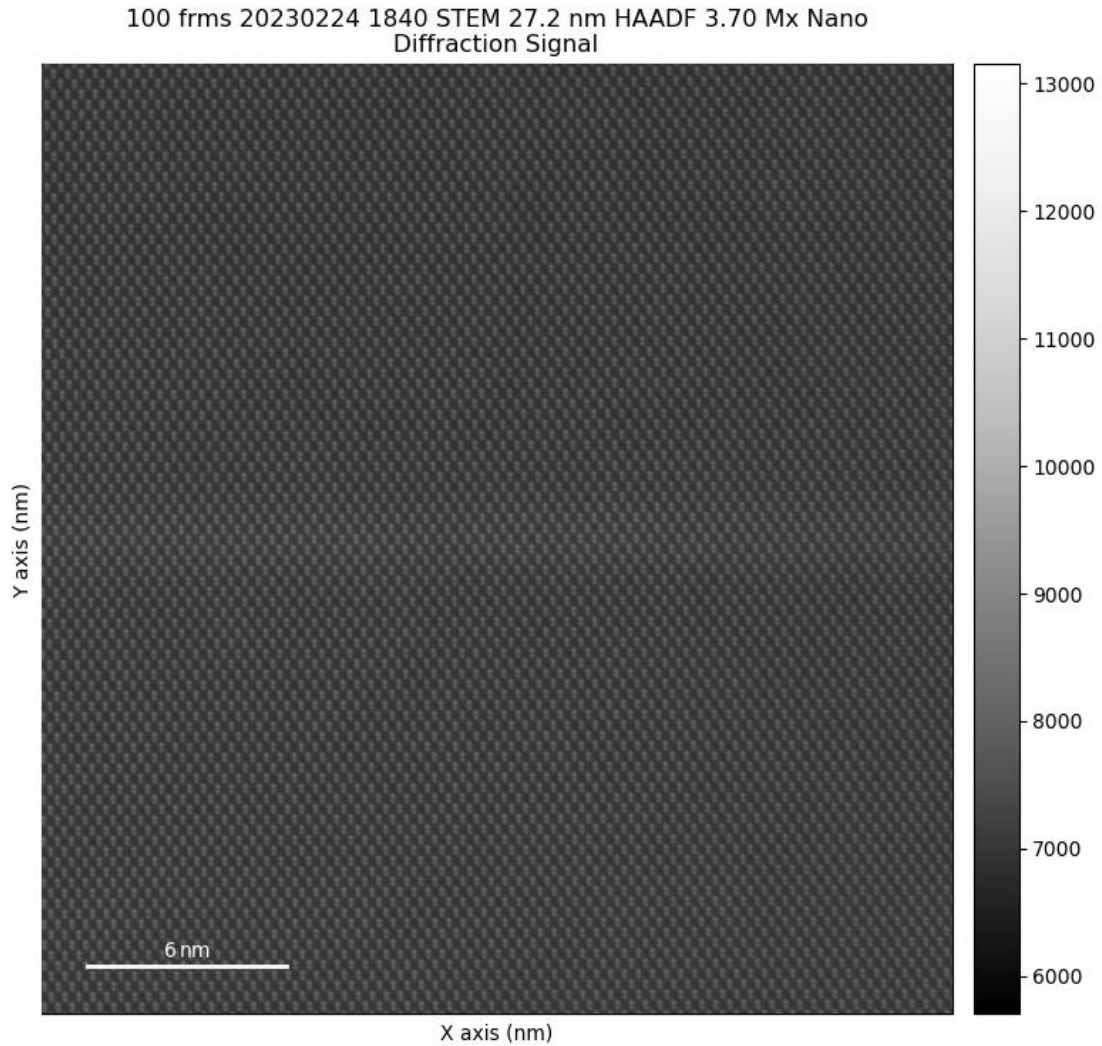


Figure 1.1. HAADF-STEM image serving as the input for the analysis. It shows a periodic atomic-column contrast and a heterostructure transition. Raw intensity is not enough to produce a quantitative composition.

1.2 Scientific question

The scientific question that this thesis addresses is: How can HAADF-STEM image data be processed and modeled in such a way that the local composition at a GaAs/AlGaAs interface can be estimated at the atomic-column scale, without overinterpreting the raw image brightness? This question can be separated into two components. The first component is a preprocessing question: how can noise be suppressed while preserving lattice information? The second component is a modeling question: how can the intensity associated with each atomic column be estimated in a more physically meaningful way than single-pixel brightness?

The answer developed in this thesis is a two-stage workflow. First, the HAADFSTEM image is transformed as if it were a noisy measurement with Poisson-like fluctuations, which is expected since the detector signal is proportional to electron counts. The Anscombe transform is explored since it can stabilize variance of Poisson-distributed data. Second, the image is modeled using a fit in which atomic-column contributions are described by localized peaks convolved with an electron probe kernel. The fitted peak volume is then extracted and used as the quantitative column descriptor.

An important term that will be used throughout this thesis is relative composition. This work does not propose measuring absolute composition from HAADF-STEM intensity alone. Rather, the analysis described herein assumes known well and barrier exists in the same image and can be used as an internal reference. Unknown interface columns are then compared with reference populations. This procedure follows logically from the practice of relative HAADF-STEM composition analysis, while respecting the practical limitation that thickness and detector geometry can prevent the analyst from determining absolute composition based on HAADF intensity alone.

1.3 Scope and contribution boundaries

This thesis contains a strict boundary around authorship. The author did not write all of the prior HAADF image-analysis workflow and did not perform all of the experimental data collection. The prior workflow is presented as prior project background. The author's adaptation of this workflow for macOS in addition to Windows, the assessment of Anscombe denoising relative to PCA-based processing for the available dataset, and authorship of the atomic-column decomposition and relative-composition code used in the final stage of the thesis were the author's contributions.

Table 1.1. Scope of authorship and technical contribution

Category	Use in thesis	Boundary wording
Group-provided HAADF-STEM data	Input data for the analysis	The microscopy data are treated as group-provided input data and are not claimed as original data collection by the author.
Prior HAADF workflow	Background workflow used to generate normalized images, PCA outputs, FFT/PSD profiles, and peak-family information.	The workflow should be described as previous-project or group work, rather than as fully original thesis authorship.
macOS/Windows adaptation	Practical contribution to running the workflow across systems.	The thesis may state that Amin Behnami adapted and ran the workflow across operating systems.
Anscombe evaluation	Thesis contribution.	The thesis should state that Anscombe-based processing performed better than PCA alone for this dataset and PSD/profile metric.
Atomic-column decomposition and relative-composition code	Main coding contribution.	The code should be described as inspired by the relative-composition literature and adapted for the GaAs/AlGaAs region of interest.
Crosstalk correction and second-pass fitting	Refinement contribution.	The correction should be described as empirical and dataset-specific, not as a universal calibration.

The composition code was authored by the thesis writer, but its theoretical basis was motivated by the HAADF-STEM literature's model-based relative-composition method. The thesis therefore does not claim that the general concept of using fitted peak volumes and known-column reference regions is novel. The contribution is the building, implementation, and interpretation of a project-specific pipeline for GaAs/AlGaAs HAADF-STEM interface analysis.

The experimental data is described as group-provided HAADF-STEM data. The sample is presented as a GaAs/AlGaAs quantum well/barrier structure with the barrier composition taken as 24% Al and 76% Ga and the well considered to be GaAs. Dr. Rosa E. Diaz Rivas directed the work and interpretation.

The latter crosstalk-correction workflow is also included as part of the author's computational contribution. This aspect of the work does not present a general universal model for crosstalk in HAADF-STEM images; rather, it enacts a project-specific empirical correction. It is based on the fitted peak volumes from the first-pass workflow for the present GaAs/AlGaAs image, to estimate the possible excess Ascolumn intensity from neighboring cation/Ga columns, subtract a modeled crosstalk image from the original ROI, and rerun the fitting workflow on the corrected

image. The final corrected output is therefore presented as a careful second-pass refinement of the relative-composition analysis, not a substitute for direct experimental validation by EELS, EDS, or a fully calibrated multislice simulation. These boundaries are made explicit as a thesis must clarify the author's own technical work from previous project infrastructure and from prior literature.

1.4 Organization of the thesis

Chapter 2 presents the physical and computational background material. The basic electron scattering and diffraction theory is summarized in Section 2.1, and the chapter ends with a summary of why HAADF-STEM can provide composition-sensitive contrast in Section 2.4. Chapter 3 provides a description of the data and image metadata as well as the preprocessing methods used to analyze it. Chapter 3 gives a description of the data and image metadata as well as the preprocessing methods used to analyze it. Chapter 4 contains the presentation of the model-based atomic-column decomposition method itself, and the mathematical formulas are provided in place of code screenshots in Appendix B. Chapter 4 also includes the first introduction of the empirical crosstalk-correction procedure that was implemented following the first-pass column fitting. Chapter 5 first reports the original fitted volume and relative-composition results, and then presents the second-pass crosstalk-corrected ROI, corrected volume-over-position trends, and updated Al-fraction estimates in Chapter 5. Chapter 6 is a discussion of the physical meaning and limitations of this correction, and Chapter 7 is a summary of the final conclusions, with validation steps still required before the corrected estimates can be interpreted as absolute composition values.

2. PHYSICAL BASIS OF HAADF-STEM INTERFACE ANALYSIS

2.1 From electron waves to diffracted signals

The starting point for the concept behind the framework used in this work was the wave nature of electrons. Electrons accelerated in an electron microscope have wavelengths that are on the scale of interatomic spacing. When an electron beam interacts with a crystalline sample, the diffracted waves can add constructively or destructively, resulting in diffraction features that are a manifestation of periodic structure. The phase difference between two electron waves can be written

$$\Delta\phi = \left(\frac{2\pi}{\lambda}\right) \cdot \Delta x \quad (2.1)$$

where λ is the electron wavelength and Δx is the path difference.

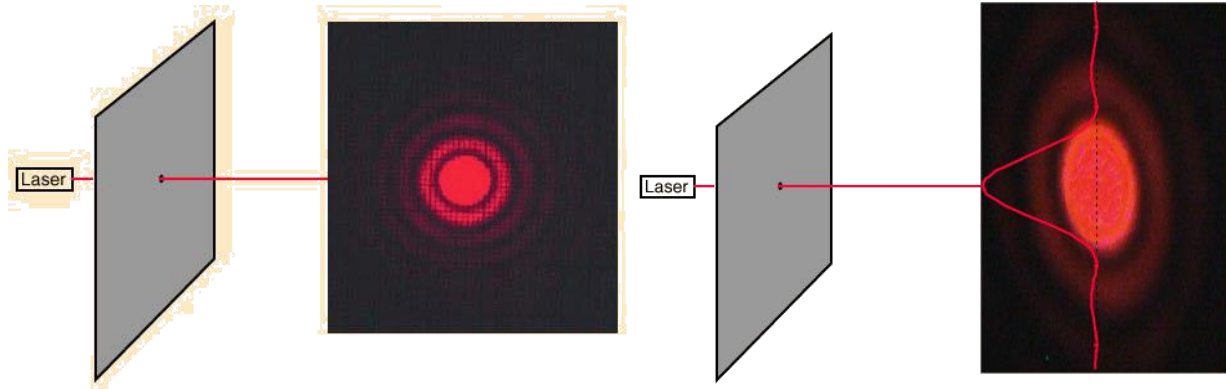


Figure 2.1. Electron-wave and diffraction concept illustrating how wave interference produces diffracted intensity from periodic structures.

This equation also illustrates why crystallinity results in structured diffraction information, rather than unstructured scattering. It also provides the conceptual bridge to Fourier analysis: if a real-space atomic lattice is periodic, it results in peaks in reciprocal space. The later PSD analysis in this thesis is based on the same principle; structural peaks in the frequency domain are interpreted as evidence that lattice information is still present after preprocessing.

The distinction between coherent diffraction contrast and incoherent high-angle scattering is a key point for this thesis. Low-angle diffraction features contain crystallographic information

but can complicate direct interpretation of composition because they strongly depend on orientation and phase. HAADF-STEM intentionally uses high-angle scattered electrons so that image intensity is more closely related to atomic number and less dominated by coherent Bragg diffraction. This is the physical reason that HAADF data were selected as the basis for column-level composition analysis.

2.2 TEM, STEM, and detector geometry

In standard TEM imaging, a wide beam is used to illuminate the sample and an image or diffraction pattern is formed from the transmitted electron wave. In STEM, the electron beam is focused to a fine probe and scanned across the sample in a raster fashion. The detector measures an intensity for each probe position. This mode of acquisition is naturally amenable to numerical image analysis, as each pixel is a measured detector response which can be directly associated with a known probe position.

Three angular quantities are used to define the experiment. The convergence semi-angle α describes the angular spread of the incident STEM probe. The scattering angle θ is dictated by the specimen, and indicates how far electrons are scattered from the optic axis. The detector collection angle β defines which scattered electrons are considered as part of the recorded signal. For HAADF-STEM, β is chosen such that only high-angle scattering is collected. As high-angle scattering scales more sensitively with atomic number, HAADF images are sometimes described as Z-contrast images.

This angular framework provided the first practical criterion for the thesis: that the image should be treated as a HAADF signal, rather than a conventional bright-field diffraction image. Once the image can be considered a high-angle intensity map, it becomes reasonable to consider whether the brightness of each column can be mapped to a relative composition descriptor. This conversion still requires preprocessing and fitting, but the physical basis is given by the detector geometry.

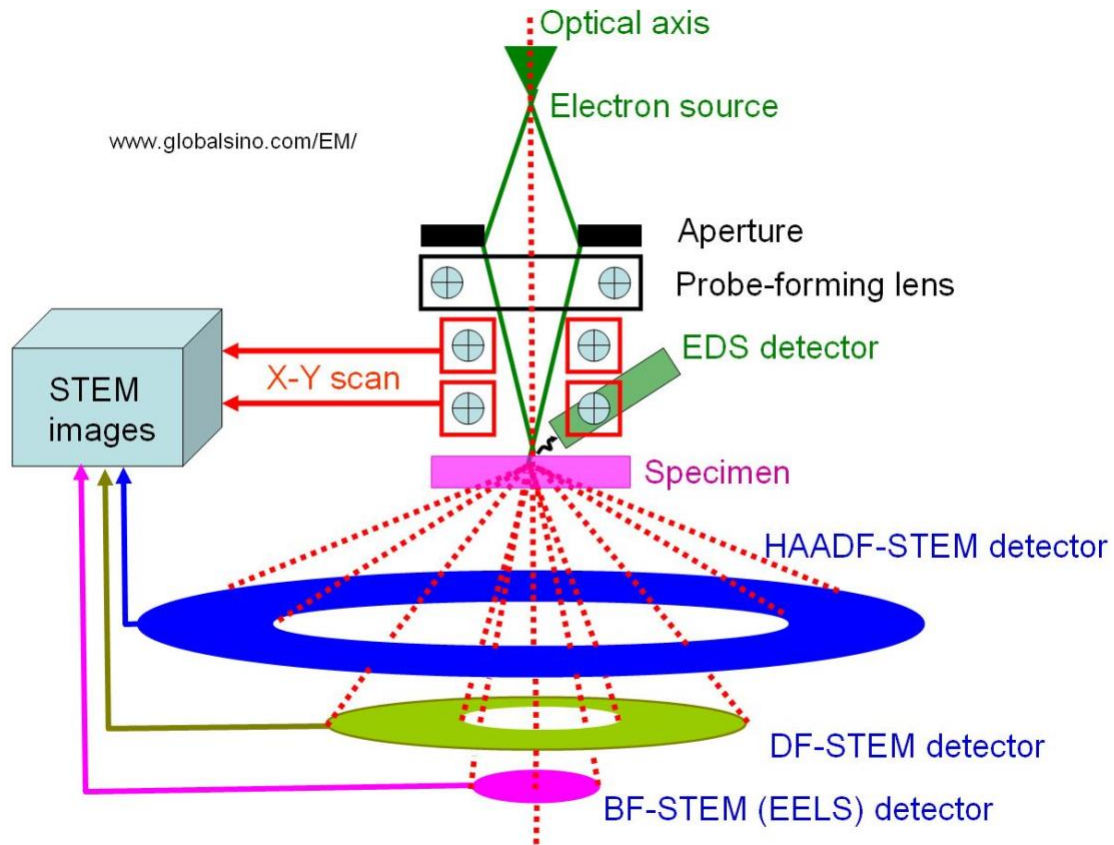


Figure 2.2. Schematic relationship between probe convergence, detector geometry, and HAADF-STEM signal collection.

Detector geometry defines which of the scattered electrons are included in the recorded HAADF-STEM signal. HAADF weighting emphasizes high-angle scattering. In the same way, the method is sensitive to the atomic number, which is useful for studying GaAs/AlGaAs interfaces. This figure ties the physical detector geometry to the intensity image that is subsequently normalized, denoised, and model-based column-decomposition fitted.

2.3 Atomic columns and Z-contrast

When the crystalline sample is oriented along a zone axis, a HAADF-STEM image can exhibit the atomic columns as localized bright spots. In such an image, each bright spot in a HAADF-STEM image corresponds to a column of atoms along the beam direction that is projected onto a single spot in the image plane. The mean intensity of a column is related to the mean atomic

number of the atoms comprising the column. While a simple qualitative statement is that heavier columns are brighter, the quantitative interpretation is more complicated and caution is required.

For a thin specimen and a fixed set of imaging conditions, the HAADF intensity can often be approximated to be a power of the atomic number. The exponent varies with the detector geometry and experimental conditions. For thicker samples, channeling, dynamical scattering, thickness variation and detector response all convolute the simple relation. For these reasons, this thesis makes use of internal references rather than claiming a direct absolute conversion from intensity to composition.

In the GaAs/AlGaAs structure imaged in this thesis, the GaAs well and the AlGaAs barrier represent two chemically meaningful reference regions, while the interface columns are the unknown region of interest. The core quantitative challenge is not just measuring if a given pixel is bright or dark, but estimating the fitted peak volume associated with each column and comparing to the fitted volumes of the known regions.

2.4 Interface chemistry

The interface in a semiconductor heterostructure can be non-ideal in a few senses. Interface roughness is a geometric perturbation to the position of the boundary from one lateral location to the next. Interdiffusion is when atoms cross the interface and mix into the adjacent layer. Segregation is when a tendency for certain atoms to preferentially partition near the boundary rather than uniformly populate the adjacent layer.

These are different physical mechanisms that could look similar on an intensity image. A smooth change in intensity over a few planes might be real intermixing, surface roughness averaged in the projection direction, or some physical interface effects combined with imaging blur. For this reason, the thesis does not claim a gradual HAADF intensity transition as one mechanism without further supporting evidence. The fitting workflow provides a quantitative description of the relative column behavior, while the physical interpretation is conservative.

The practical consequence is that the analysis must report both the numerical output and the limitations of the inference. A fitted column classified as intermediate does not automatically imply a unique atomic mechanism. It indicates that, with the model and reference assumptions, the column's fitted peak volume is between or overlapping the reference populations. This conservative interpretation is carried throughout the results chapter.

3. DATA, PREPROCESSING, AND SPECTRAL EVALUATION

3.1 Material system and image metadata

The material system addressed in this thesis is the GaAs/AlGaAs quantum well/barrier structure. The barrier and well material was assumed to be $\text{Al}_{0.24}\text{Ga}_{0.76}\text{As}$ and GaAs, respectively. The HAADF-STEM image data was acquired prior to the current thesis work and is used in this work as acquired for a group effort under the guidance of Dr. Rosa E. Diaz Rivas. The available metadata provide the numerical scale and experimental context for the work, and the relevant values are summarized in Table 3.1.

These values are used for axis scaling, pixel dimension interpretation and physically motivated probe model construction. Some of the optical parameters required for the ideal literature model were not available or were themselves treated as assumed parameters. This is an important limitation and is discussed subsequently.

Table 3.1. Imaging and sample metadata used as the computational basis for the thesis

Parameter	Value used in Thesis
STEM magnification	3.67 Mx
Field of view	27.24 nm
Image size	4096 x 4096 pixels
Pixel size	6.652 pm
Dwell time	50.0 ns
Camera length	37 mm
Accelerating voltage	300 kV
Beam convergence	17.9 mrad
Beam current	116 pA
Defocus	193 nm
Barrier composition	Al _{0.24} Ga _{0.76} As (24% Al, 76% Ga)
Well composition	GaAs

3.2 Practical motivation for relative quantification under limited access to complementary measurements

In this thesis the sample and data analysis were restricted to what was experimentally accessible due to limitations in the available microscopy data and scarce access to repeated instrument sessions and complementary composition-calibrated measurements. A fully calibrated composition measurement at every interface column would consist of HAADF-STEM imaging with additional spectroscopic data (e.g. electron energy-loss spectroscopy or EDS) as well as repeated measurements at multiple locations under controlled imaging conditions. In practice, instrument time, operating cost, sample availability, and access to complementary measurements can constrain the additional data that can be collected for a thesis-scale project. These considerations provide the motivation for the relative quantification framework described in this thesis. Rather than attempting to claim an absolute composition measurement from raw HAADF-

STEM brightness, in this thesis, fitted atomic-column quantities are compared between regions of known composition and unknown interface columns within the same image. The GaAs quantum well and the AlGaAs barrier act as internal references to allow interpretation of the interface columns relative to the nearby known regions, reducing the dependence on external calibration and making use of the information available in the HAADF-STEM image while still maintaining scientific caution about the limitations of an intensity-based composition analysis. The resulting composition values should therefore be interpreted as relative estimates resulting from a model-based analysis rather than independent, calibrated absolute chemical measurements.

3.3 Normalization and initial image handling

The first step of the HAADF workflow is to load the heterostructure image and apply spatial scaling. This is followed by detector-related normalization. The need for normalization comes from the requirement of quantitative comparison of intensities after accounting for a background detector response which should be subtracted and scaling by the incident beam or detector reference (if available). In a simple form, the normalized intensity can be expressed as

$$I_{\text{norm}} = (I_{\text{exp}} - I_{\text{bkg}}) / (I_{\text{beam}} - I_{\text{bkg}}) \quad (3.1)$$

where I_{exp} is the measured image intensity, I_{bkg} is a background or dark reference, and I_{beam} is the incident-beam or detector reference signal. The output of this stage is a normalized image, which is a more suitable representation for comparison of intensity variations across the field of view.

The workflow also crops the boundary pixels to reduce the edge artifacts associated with the drift correction and image processing. This step matters because the Fourier-domain evaluation and peak fitting are sensitive to artificial discontinuities near the image borders. An edge artifact may produce frequency-domain power that is unrelated to the crystal lattice and could be misinterpreted as structural signal.

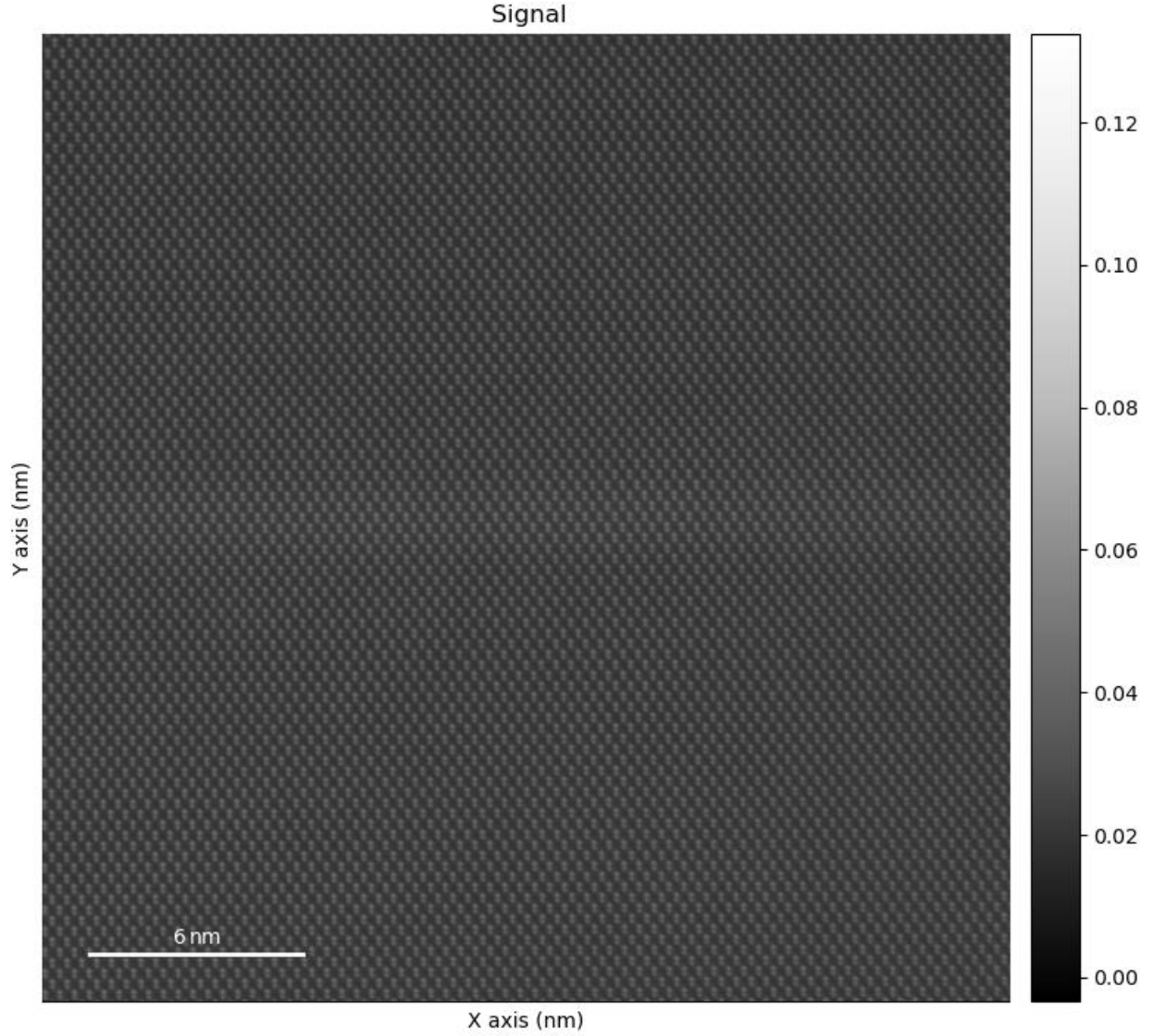


Figure 3.1. Normalized HAADF-STEM image after detector-related normalization and cropping. This image is the basis for subsequent denoising and frequency-domain comparison.

3.4 Noise model and variance stabilization

STEM image intensity is a proxy for the number of electrons detected at each probe position. Because electron detection is a counting process, the noise can be well-modeled as Poisson-like under many conditions. For a Poisson random variable X with mean μ , the variance is also μ :

$$X \sim \text{Poisson}(\mu), E[X] = \mu, \text{Var}(X) = \mu \quad (3.2)$$

This property means that brighter regions have larger absolute fluctuations. A denoising method that assumes constant-variance Gaussian noise may not behave optimally on raw Poisson-like data. The Anscombe transform addresses this problem by stabilizing the variance:

$$A(X) = 2 \left(\sqrt{X + \frac{3}{8}} \right) \quad (3.3)$$

After the transform, the noise is approximately constant-variance for sufficiently large counts. The inverse transform returns the image to the original intensity domain after denoising. The thesis uses the standard Anscombe logic as a preprocessing strategy and compares the resulting image behavior with PCA-based processing. The comparison is made using image-domain interpretation and Fourier-domain radial profiles, not by visual smoothness alone. The point of the denoising stage is not to make an image aesthetically cleaner. The point is to preserve the atomic-lattice information that will later be required for column detection and fitting. Therefore, the evaluation criterion must ask whether structural peaks remain present in frequency space while the high-frequency noise tail is reduced.

3.5 Fourier transform and radial PSD evaluation

A periodic atomic lattice results in regular peaks in the Fourier transform of the image. Suppose $f(x,y)$ is the image intensity. The two-dimensional Fourier transform is

$$F(u,v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) e^{-2\pi i(ux+vy)} dx dy \quad (3.4)$$

The power spectral density is proportional to the squared modulus of the Fourier transform:

$$\text{PSD}(u,v) = |F(u,v)|^2 \quad (3.5)$$

Radial averaging converts the two-dimensional frequency representation into a one-dimensional profile as a function of spatial frequency. This profile can be used to compare raw, PCA-denoised, and Anscombe-processed images. Peaks at low and intermediate spatial frequencies correspond to structural periodicities. A flat or slowly varying tail at higher frequencies is often associated with noise and residual background. Using a radial PSD/profile is a scientifically appropriate metric because it relates the preprocessing step to the physical lattice

information used later. If a method reduces the high-frequency tail but also suppresses the structural peaks, it could be detrimental to the composition analysis. If it preserves the structural peaks while reducing the tail, it is a preferable preprocessing choice for this dataset.

3.6 PCA-based denoising as a comparison method

Principal component analysis decomposes image data into a set of orthogonal components, which are ordered by variance. The denoising workflow preserves high-variance components and discards the low-variance ones. The reconstructed image can thus be written schematically as

$$D_k = \sum_{i=1}^k t_i p_i^T \quad (3.6)$$

where D_k is the rank- k reconstruction, t_i are score vectors and p_i are component vectors. PCA's utility is that it can simultaneously suppress noise and retain dominant image structure. The limitation is that PCA organizes information by variance, not physical meaning. Noise contributes to variance, and weak structural signals can be lost if too few components are retained.

The HAADF workflow includes reconstructions with several component counts, and identifies the corresponding PSD peaks. A peak-attenuation table from the earlier analysis indicates that different spatial-frequency families are affected differently by the number of retained PCA components. This observation further motivates comparing denoising methods with structural metrics instead of just visual impression.

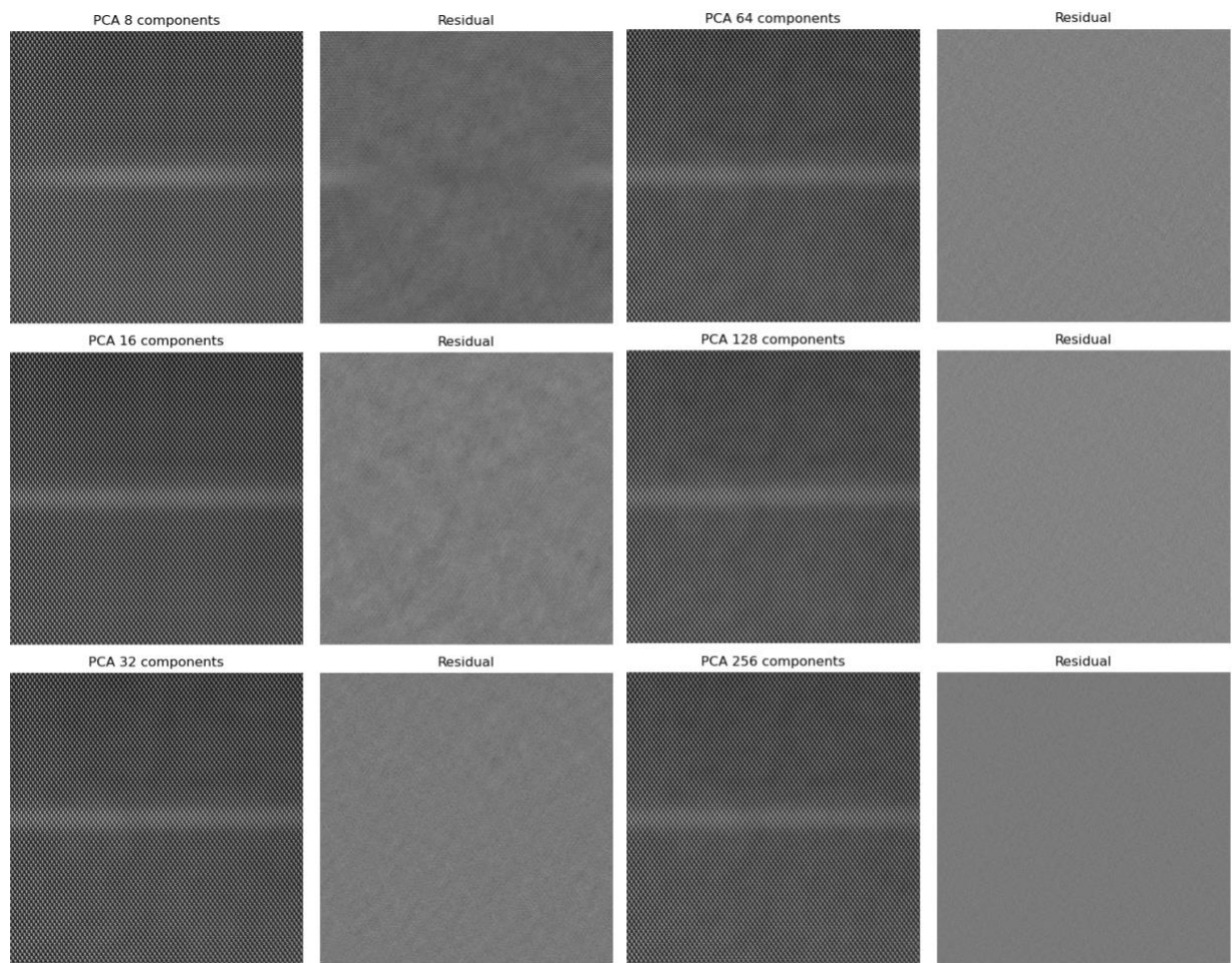


Figure 3.2. PCA reconstruction and residuals. PCA reconstruction and residual panels showing how component count changes the retained lattice contrast and residual structure.

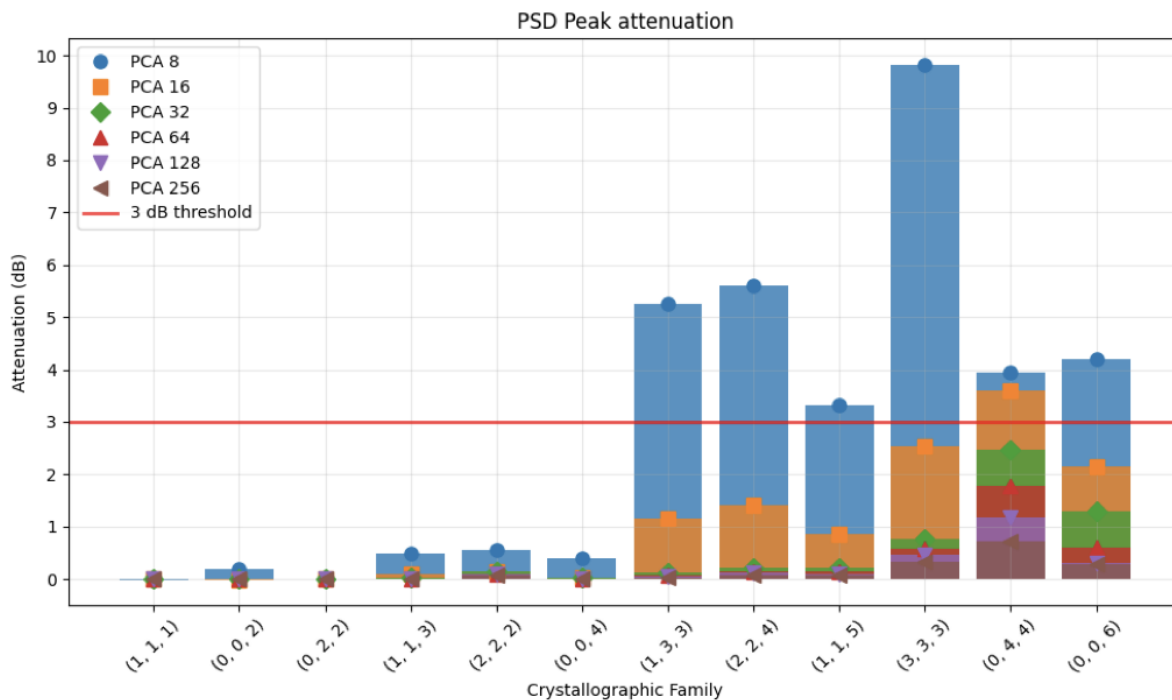


Figure 3.3. PSD peak attenuation by crystallographic family. Attenuation of PSD peak families for different PCA component counts. The figure shows that component selection affects crystallographic peak families differently.

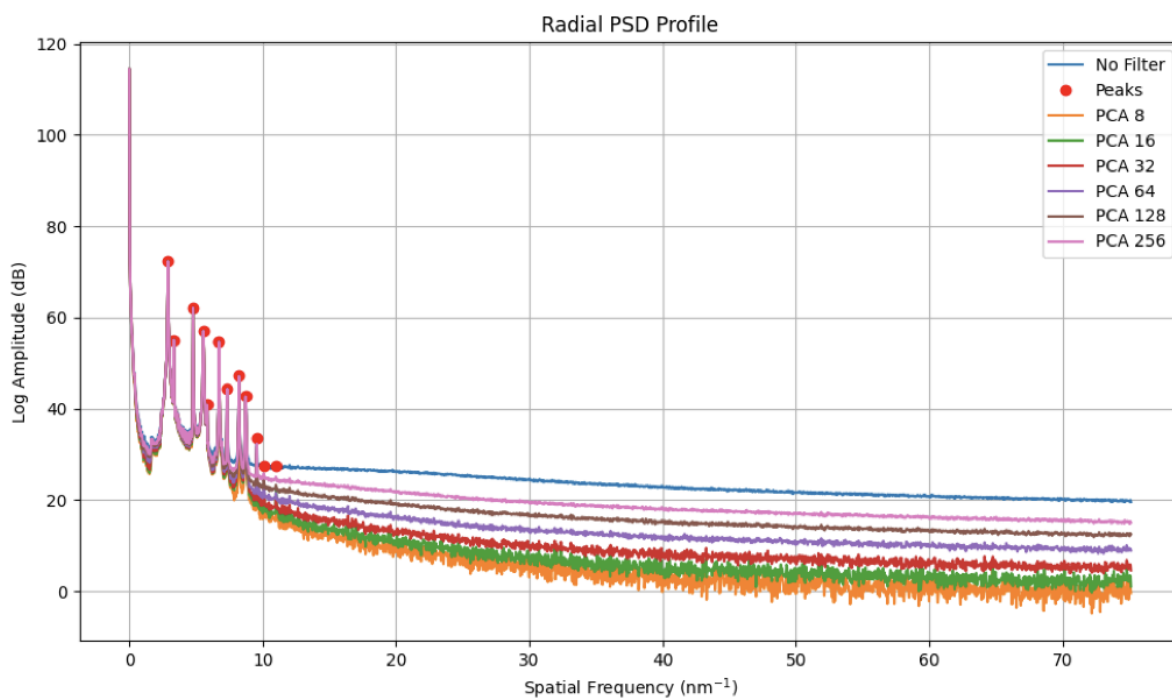


Figure 3.4. Radial PSD profile. Radial PSD profile used to connect the Fourier-domain evaluation to structural peaks in the HAADF-STEM data.

3.7 Anscombe-based processing and observed output

The Anscombe-based workflow includes a variance stabilization step prior to denoising. The basic idea is to apply a transform that turns the Poisson-like noise into approximately constant-variance, then smooth/smooth-related (PCA) in a domain where the noise is approximately uniform, then back-transform. The transform/denoising sequence in basic notation is

$$\text{image} \rightarrow A(\text{image}) \rightarrow \text{denoise}[A(\text{image})] \rightarrow A^{\{-1\}}\{\text{denoise}[A(\text{image})]\}$$

The denoising comparison output illustrates the practical motivation for using this approach. In the PCA-only profile, structural peaks are visible but the high-frequency tail is elevated. In the Anscombe-based result, the high-frequency tail is more strongly reduced while the main structural peaks remain visible. The result supports the careful conclusion that Anscombe-based preprocessing performed better than PCA alone for this dataset and this PSD/profile metric.

The claim is intentionally circumscribed. The thesis does not claim that Anscombe is generally superior for all STEM images. The evidence supports a narrower and more scientifically defensible statement: for the available GaAs/AlGaAs HAADF-STEM dataset, the Anscombe-based workflow reduced the noise-associated tail intensity while preserving the lower-frequency structural peaks needed for atomic-column analysis.

4. MODEL-BASED ATOMIC-COLUMN DECOMPOSITION

4.1 Reason for moving beyond raw brightness

After pre-processing, the analysis must deconvolve an image into column-specific quantities. Raw pixel brightness is not a good descriptor of composition, because each atomic column's intensity is spread over several pixels and adjacent columns overlap. The measured image is also blurred by the finite shape of the probe. A pixel in the center of one column could receive signal from nearby columns and background. The analysis therefore estimates fitted peak volume instead of measuring single-pixel intensity.

The approach follows the conceptual direction of model-based HAADF-STEM relative-composition analysis. The object is modelled as a set of localized atomic-column contributions. The object is convolved with a probe/source kernel to produce a predicted image. The unknown parameters are then adjusted so that the predicted image matches the measured ROI. This turns the problem from visual inspection into parameter estimation.

4.2 Image formation model

We can write the fitted object model as a constant background plus a sum of Gaussian-like atomic-column peaks:

$$O(x, y) = b + \sum_i a_i \exp\left[\frac{-(x-x_i)^2 + (y-y_i)^2}{2\sigma^2}\right] \quad (4.1)$$

where b is the background term, a_i is the amplitude of column i , (x_i, y_i) is the column center, and σ is a shared width parameter. The predicted HAADF image is the object convolved with a probe/source kernel K :

$$I_{\text{pred}}(x, y) = [O * K](x, y) \quad (4.2)$$

The measured ROI is denoted I_{meas} . The fitting problem minimizes the pixel-wise residual between predicted and measured intensity:

$$\min \sum_{x, y} [I_{\text{pred}}(x, y) - I_{\text{meas}}(x, y)]^2 \quad (4.3)$$

This least-squares formulation encodes the key research decision: the image is not regarded as an ensemble of uncorrelated pixels, but as the product of an image-formation process. The inclusion of overlapping peaks and a background term means that the fitted parameters are more useful for column comparison than the raw pixel values.

4.3 Probe kernel and relativistic wavelength

The probe kernel is parametrized by microscope parameters. The relativistic electron wavelength is needed because the microscope voltage sets the wavelength of the electron beam. A standard expression for the relativistic wavelength is

$$\lambda = \frac{h}{\sqrt{2meV\left[1+\frac{eV}{2mc^2}\right]}} \quad (4.4)$$

where h is Planck's constant, m_e is the electron rest mass, e is the elementary charge, V is accelerating voltage, and c is the speed of light. The aperture and aberration terms determine a probe intensity distribution, to which source broadening is convolved to form the final kernel. In the implemented workflow, the kernel is normalized so that convolution conserves total intensity. Not all microscope aberration coefficients were known with full certainty. The probe model should be viewed as a physically-motivated approximation rather than a fully calibrated microscope simulation. The effect of this on the precision of absolute values is unclear, but internal relative comparison should still be possible if the same model is applied consistently to the well, barrier, and interface regions.

4.4 Interface localization and ROI selection

The first computational step after denoising is interface localization. The image is row-averaged, producing a one-dimensional intensity profile. The smoothed row mean exposes the coarse transition between the well and barrier regions. The derivative of the smoothed row profile is then used to find positions where the intensity changes abruptly. These positions are used as candidate interface locations.

Once the interfaces are identified, a region of interest is selected in the vicinity of the interfaces. This step is required for computational efficiency as well as scientific interpretation.

Fitting the full 4096 x 4096 image would be wasteful and unnecessary for interface characterization. A local ROI contains known columns on either side of the interface. Unknown interface columns can then be compared against their immediate neighbors under similar imaging conditions.

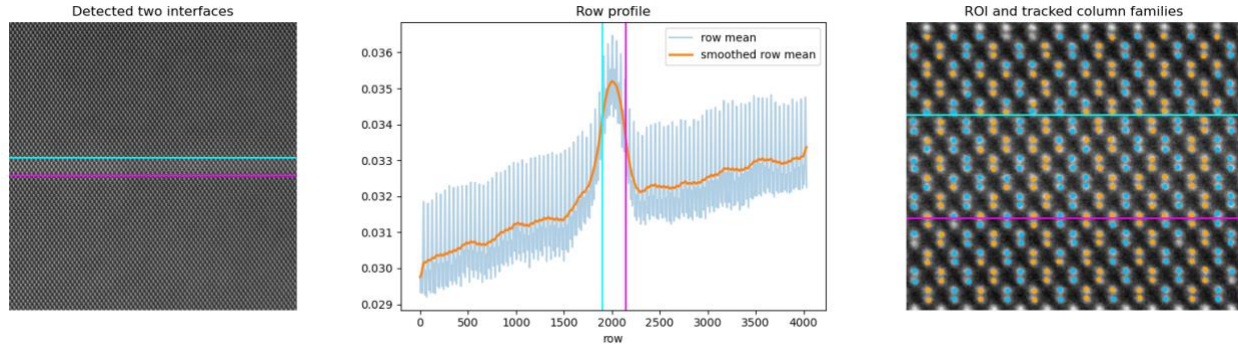


Figure 4.1. Interface localization, row-profile behavior, and selected ROI used to define the upper and lower interface analysis regions.

4.5 Column detection, family selection, and plane grouping

Candidate atomic columns are detected as local maxima in the ROI. Because the atomic lattice is periodic, an autocorrelation-based estimate of lattice spacing is used to choose peak-detection parameters such as minimum peak spacing and plane tolerance. This reduces arbitrary tuning and connects the detection settings to the image itself.

After peaks are detected, they are grouped into planes based on row position. Alternating column families can appear in the projected lattice, so the analysis separates columns into families and chooses a common family for the upper and lower interfaces. The output indicated that column family 0 was selected for both interfaces. Selecting one family consistently avoids comparing physically different projected column types as if they were equivalent.

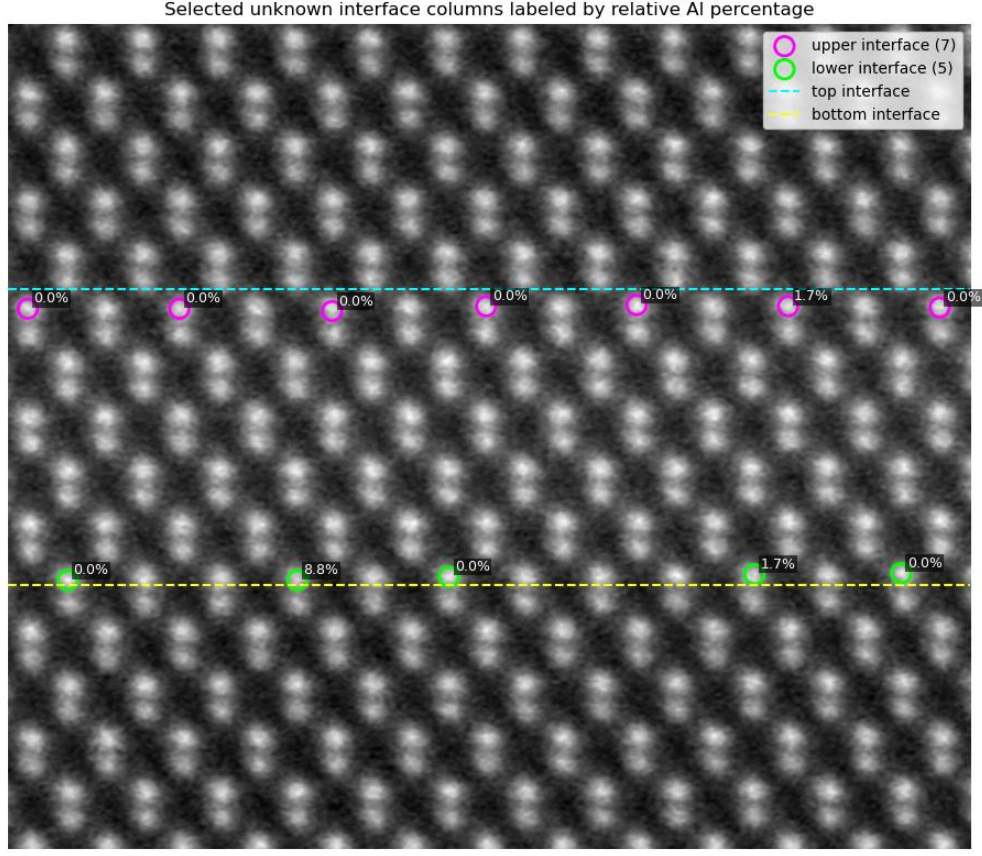


Figure 4.2. Detected upper and lower unknown interface columns with interface markers and selected column positions.

4.6 Peak volume as the column quantity

As just discussed, the relevant quantity after the fit is not amplitude alone. For a two-dimensional Gaussian peak, the integrated volume is proportional to amplitude times width squared:

$$V_i = 2\pi a_i \sigma^2 \quad (4.5)$$

where V_i is the estimated peak volume of column i . This expression is important to the thesis because it transforms the fitted local peak into an integrated column descriptor. It is closer to total scattered intensity than a single-pixel value and less sensitive to the exact pixel position of the peak center.

The use of peak volume is also consistent with the logic of the relative-composition literature: known columns provide distributions of fitted volumes, and unknown interface columns

are compared to those distributions. The method is statistical because real columns of the same type do not produce identical estimates. Counting noise, scanning instability, local defects, and model mismatch all introduce variation.

4.7 Tolerance intervals and classification

To classify unknown columns, reference columns are summarized with tolerance intervals. A two-sided tolerance interval for approximately normally distributed values in a reference set can be expressed as

$$\mu \pm k_s \quad (4.6)$$

where μ is the sample mean, s is the sample standard deviation, and k depends on the desired population coverage p and confidence g . In the implemented workflow

$p = 0.90$ and $g = 0.90$ were used. The interval does not create a hard universal composition boundary; rather, it provides a statistically principled way to compare unknown columns to known reference columns from the same image.

Because well and barrier tolerance intervals may overlap, classification is not always unambiguous. An unknown column may be well-like, barrier-like, overlapping/ambiguous, or outside the expected intervals. The presence of ambiguity is not a failure of the method; rather, it is an honest representation of uncertainty that is caused by overlapping reference distributions or by an interface column that falls in a mixed/noisy region.

4.8 Empirical crosstalk correction and second-pass analysis

In addition to first-pass peak-volume fitting and tolerance-interval comparison, one more correction has been applied to account for potential crosstalk between adjacent atomic columns. Crosstalk is, as defined in this thesis, intensity that is recorded at the projected position of one atomic column but is partially dependent on the signal contribution from neighboring columns. This effect can be important because finite probe shape, peak overlap, and the convolution process means that each fitted column is not strictly isolated from neighboring columns. If the contribution is not accounted for, some of the measured intensity change near the interface could be misattributed to a local change in composition.

The correction was implemented as a second-pass computational workflow. The first pass used the original denoised image and the model-based fitting procedure described above. The result of the first pass was that each fitted column had a row number, atom-family label, a region assignment, and an estimated peak volume. These rows were then sorted into simple reference regions. Barrier rows were assigned to known AlGaAs reference rows, well rows were assigned to known GaAs reference rows, and the other rows were either intermediate or other rows. This created reference populations for both As columns and cation/Ga columns.

The empirical assumption used for this correction is that a change in the As-column fitted volume between the barrier reference and the well reference is not assumed to be a true increase in specimen thickness. The excess is instead assumed to be crosstalk from nearby cation/Ga columns. This is a conservative assumption because the As sublattice is not the intended composition variable in the GaAs/AlGaAs comparison. The cation sublattice is the sublattice that is really changing chemically, because the well is being treated as GaAs and the barrier is being treated as Al_{0.24}Ga_{0.76}As. Therefore, an apparent difference in As intensity between the well and barrier regions can be used as an empirical measure of how much neighboring cation intensity leaks into each As-column measurement.

The first-pass fitted volumes were grouped into atom family and reference region. The average fitted volume of As columns in the barrier was compared to the average fitted volume of As columns in the well. The difference was the excess As volume,

$$V_{\text{excess,As}} = V_{\text{As,well}} - V_{\text{As,barrier}} \quad (4.7)$$

This excess was then divided by the average fitted cation/Ga volume in the well and by the number of neighboring cation columns that are used in the local correction. The number of neighboring cation/Ga used in the implemented code is 3. The empirical cation to As crosstalk coefficient is then given by

$$k_{\text{cat} \rightarrow \text{As}} = V_{\text{excess,As}} / (3 V_{\text{cat,well}}) \quad (4.8)$$

This is not a universal microscope parameter. It is an empirical coefficient that is calculated from this dataset and this ROI. Its purpose is only to estimate how much of the As-column fitted volume might be explained by nearby cation/Ga columns under the same imaging and fitting conditions.

For each As column, the code finds nearby cation/Ga columns from adjacent rows and selects the three nearest cation/Ga neighbors in column position. The fitted volumes of those neighboring cation/Ga columns are then summed. The estimated crosstalk contribution to each As column is then calculated as

$$V_{\text{crosstalk,As}} = k_{\text{cat}} \rightarrow \text{As} \sum(V_{\text{cat,neighbors}}) \quad (4.9)$$

This column-level crosstalk estimate is then converted into an image-level correction. Because the original fitted object represented atomic-column peaks using Gaussian-like functions, the estimated crosstalk volume must be converted back into a Gaussian peak amplitude. For a two-dimensional Gaussian peak, the relationship between peak volume and amplitude is $V = 2\pi a_i \sigma^2$. The crosstalk amplitude is therefore estimated as

$$a_{\text{crosstalk}} = V_{\text{crosstalk,As}} / (2\pi\sigma^2) \quad (4.10)$$

Gaussian crosstalk peaks are then placed at the As-column positions and convolved with the same probe/source kernel that was used in the original model. This creates an estimated crosstalk-intensity image. The corrected ROI is given by subtracting the estimated crosstalk intensity from the original ROI,

$$I_{\text{corrected}} = I_{\text{measured}} - I_{\text{crosstalk}} \quad (4.11)$$

Negative intensity values are clipped to zero, to prevent nonphysical negative image intensity. This corrected ROI is then placed back into the image array and is used as the input for a second run of the same column-detection, fitting, and output workflow.

This means that the correction was not applied only as a numerical postprocessing step to the final table. Instead, it created a corrected intensity image and allowed the entire model-based fitting pipeline to be repeated.

The final stage of this second-pass workflow was to estimate relative Al fraction using a local cation/As ratio. For each cation/Ga column, nearby As columns were identified and averaged to create a local As reference. The local cation/As ratio was then computed as

$$R_{\text{cat/As}} = V_{\text{cat}} / V_{\text{As,local}} \quad (4.12)$$

The ratio in the GaAs well is treated as the $x = 0$ reference, and the ratio in the AlGaAs barrier is treated as the $x = 0.24$ reference. A relative Al-fraction estimate was then calculated by linearly mapping the local ratio between these two references, $x_{Al} = 0.24 (R_{well} - R_{cat/As}) / (R_{well} - R_{barrier})$

Values outside of the physical range were clipped between 0 and 0.24 for the plotted corrected profile. This ratio-based result should be interpreted as an internal relative estimate. It reduces sensitivity to shared local intensity variation because each cation column is being compared to nearby As columns, but it still depends on the assumptions of local thickness similarity, correct reference-region assignment, and validity of the empirical crosstalk correction.

5. COMPUTATIONAL OUTPUTS AND INTERPRETATION

5.1 HAADF workflow outputs

The previous HAADF workflow generated a series of outputs, which now serve as the starting point of this thesis's preprocessing stage: normalized images, PCA reconstructions, FFT/PSD profiles, peak-family tables and attenuation plots. These outputs are provided as evidence of analysis here, rather than screenshots of the code in use. Together they illustrate how preprocessing alters the spatial-frequency content of the image.

The normalized image itself establishes that the heterostructure is visually discernible, and that atomic-column periodicity is present. The PSD outputs then quantify this structural content. The PCA attenuation outputs demonstrate that component selection affects different peak families differently and thus motivates the evaluation of Anscombe-based processing as a more appropriate noise-stabilization stage for this dataset.

5.2 Anscombe vs. PCA output

The output image of the denoising comparison shows that the Anscombe-based processing suppressed the high-frequency tail more than the PCA alone, while keeping the lower frequency peaks from the lattice structure. The structural peaks must be retained because the later decomposition stage relies on accurate detection and fitting of the atomic columns.

The interpretation of this result is therefore: The Anscombe-based processing performed better for this dataset and for this PSD/profile metric. It reduced the high-frequency content that was not part of the physical signal, and it kept the actual intensity variation from the structure. This result supports using the Anscombe-processed image as the input for the model-based decomposition stage.

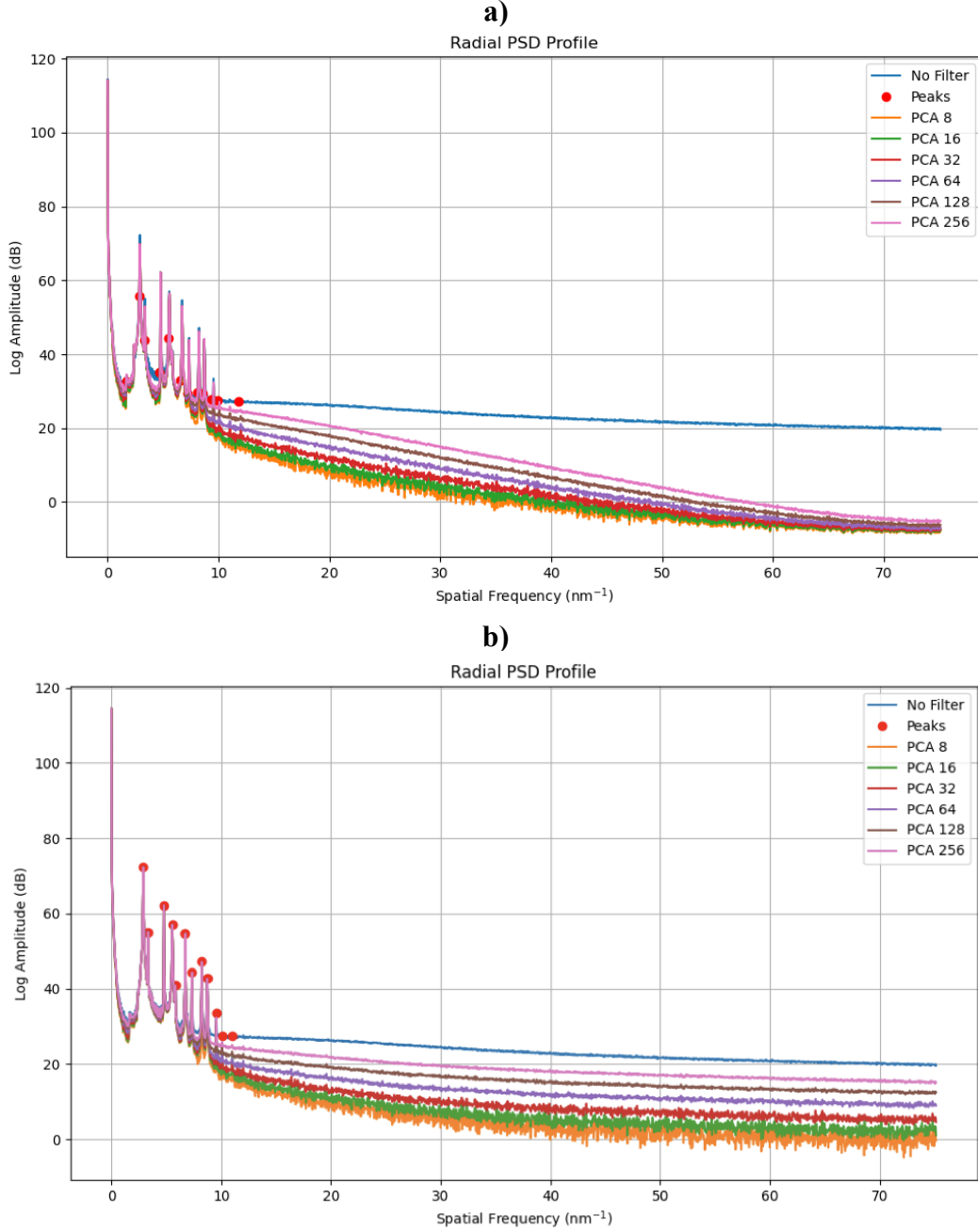


Figure 5.1. a) Anscombe vs. b) PCA radial PSD comparison. Radial PSD comparison between PCA-only processing and Anscombe-based preprocessing. The Anscombe-based workflow suppresses the high-frequency tail more strongly while preserving lower-frequency lattice peaks needed for column fitting.

5.3 Interface detection and region assignment

The decomposition pipeline identified two interfaces for the upper and lower transitions of the quantum well/barrier structure. The row profile output includes a smoothed intensity trend and

interface markers. The ROI view shows tracked column families and region boundaries. This output is significant because relative composition analysis requires well-defined assignments of known barrier references, known well references, and unknown interface columns.

The output also makes a practical point: the upper and lower interfaces should not be constrained to behave identically. They are analyzed separately, but the same selected column family is used so that the comparison remains consistent. This approach allows asymmetry between interfaces to be revealed in the results while avoiding family-dependent bias.

5.4 Fitted peak-volume summary

The last interface-analysis output selected column family 0 for both interfaces. The reference and unknown groups gave the summary values in Table 5.1. The well reference mean peak volume is 9.223 for both interface comparisons, because the same well reference was used. The upper barrier reference mean is 7.622, and the lower barrier reference mean is 7.226. The upper interface contains 7 unknown columns, while the lower interface contains 5 unknown columns.

These values indicate that the fitted well reference is brighter than the fitted barrier reference under the chosen family and ROI conditions. This is consistent with the idea that Ga-rich columns can create stronger HAADF intensity than Al-containing barrier columns, but the thesis does not make this a direct conversion into absolute composition without calibration and thickness correction.

Table 5.1. Fitted peak-volume summary for selected column family 0

Quantity	Upper interface	Lower interface
Barrier reference region	top barrier ref	bottom barrier ref
Unknown region	top interface unknown	bottom interface unknown
Selected family	0	0
Mean barrier volume	7.622	7.226
Mean well volume	9.223	9.223
Barrier reference columns	35	34
Well reference columns	30	30
Unknown columns	7	5

5.5 Tolerance-interval results

The upper interface has a barrier tolerance interval of approximately 5.419 to 9.825 and a well tolerance interval of approximately 7.119 to 11.328. The lower interface has a barrier tolerance interval of approximately 4.161 to 10.290 and a well tolerance interval of approximately 7.119 to 11.328. The intervals overlap a good bit, even more for the lower interface.

The fact that the intervals overlap is an honest and important result. It means that many of the unknown columns cannot be uniquely classified as barrier-like or well-like with these tolerance intervals. Several unknown columns lie in an ambiguous/overlapping region. Hence the output should be considered a relative statistical comparison, not a definitive atomic species assignment.

The classification plots convey this. Orange points are barrier references, blue points are well references, and unknown interface columns are plotted relative to shaded tolerance bands. The interface marker shows where the unknown region is in space. The figure shows that the method generates interpretable output, but also why we must report uncertainty.

Table 5.2. Tolerance-interval classification summary

Interface	Barrier tolerance interval	Well tolerance interval	Interpretation
Upper interface	5.419 to 9.825	7.119 to 11.328	Substantial overlap is present; several unknown columns are ambiguous.
Lower interface	4.161 to 10.290	7.119 to 11.328	The overlap is larger; classification uncertainty must be reported.

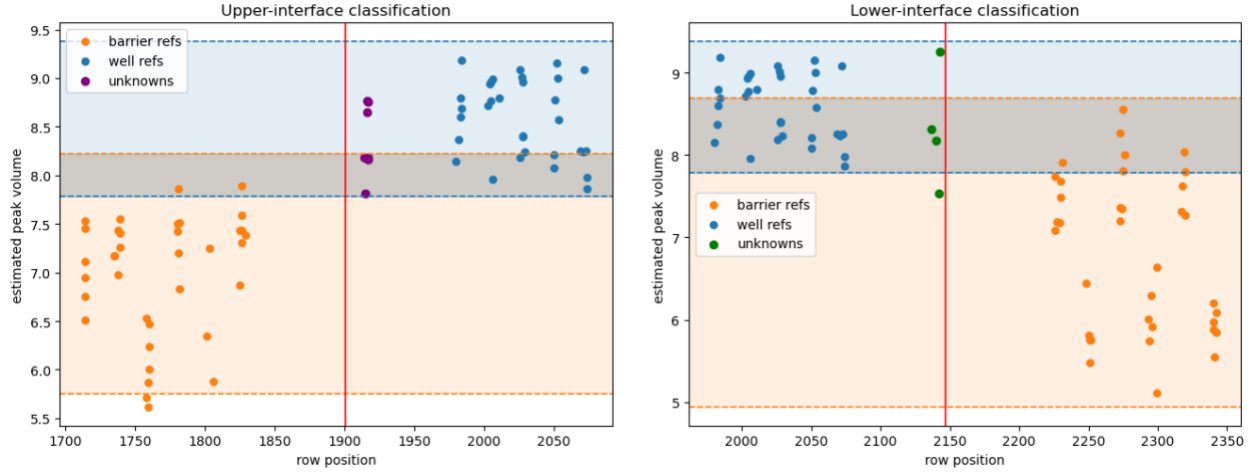


Figure 5.2. First-pass interface classification. The output should be interpreted cautiously because reference tolerance intervals overlap.

5.6 Relative Al-fraction output

The final output is estimated relative Al-fractions for the unknown interface columns. The relative estimates are made by mapping the fitted unknown-column peak volumes onto the well and barrier references. Conceptually, given a fitted peak volume V_{unknown} of an interface column, and the well reference volume V_{well} and barrier reference V_{barrier} , a relative barrier fraction can be written:

$$f_{\text{barrier}} = (V_{\text{well}} - V_{\text{unknown}}) / (V_{\text{well}} - V_{\text{barrier}}) \quad (5.1)$$

and the Al fraction is then scaled by the known barrier Al content:

$$x_{\text{Al}} = 0.24 f_{\text{barrier}} \quad (5.2)$$

This mapping assumes that the reference values are valid and that the well-to-barrier changes in intensity can be approximately linearized for relative comparison. Because the tolerance intervals overlap and the thickness is not independently corrected, the resulting Al fractions should be considered preliminary relative estimates. The output is good for visualizing trends and identifying ambiguous columns, but should not be reported as an independently validated absolute composition map.

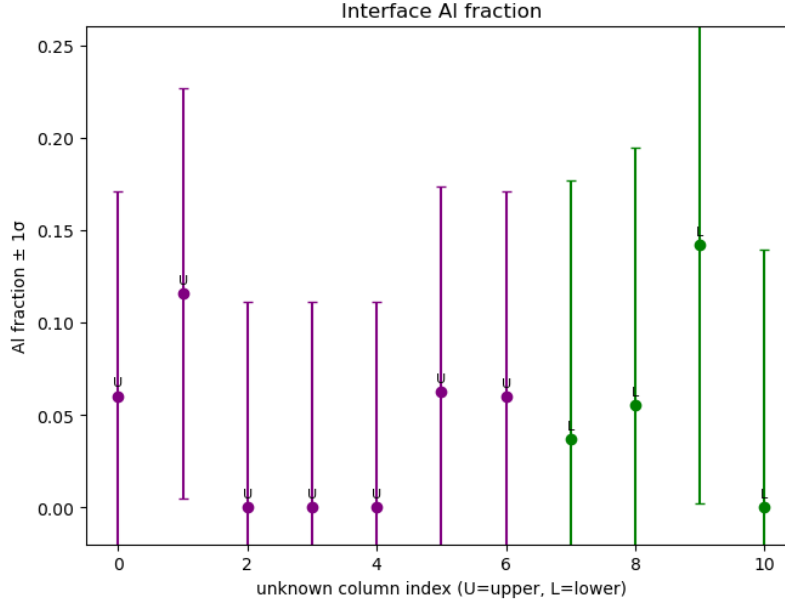


Figure 5.3. First-pass relative Al-fraction output.

5.7 Interpretation of the code outputs

The outputs of the code show that the pipeline can go from a denoised HAADF image, through interface localization, column detection, fitted peak volumes, reference statistics, classification and relative Al-fraction estimates. This is the primary technical contribution of the thesis. The output has not yet established a final physical mechanism for interface broadening. Instead, the work has created a reproducible computational framework for quantitatively and cautiously reporting on local composition.

The most significant scientific finding is that the current reference distributions are broad enough to cause classification ambiguity. This may arise from noise, sample thickness variation, column-family effects, incomplete probe calibration, or real interface mixing. The output therefore supports both the need for uncertainty reporting and future validation. A refined scientific conclusion must report this limitation, rather than obscure it.

5.8 Empirical crosstalk-correction output

After producing the first-pass relative-composition output, the analysis was broadened by an empirical construction of crosstalk correction. The intent of this step was to see if some portion of the apparent As-column intensity difference between well and barrier regions could be

explained by neighboring cation/Ga-column signal. It is worth repeating that the first-pass output already showed broad reference distributions and overlapping tolerance intervals. The crosstalk correction therefore was a second-pass refinement of the output rather than an assertion of an absolute/compositional calibration.

The first-pass fitted-volume table was regrouped by row number, atom family, and reference region. The output then produced separate row-averaged fitted volumes for cation/Ga columns and As columns over the full ROI. The As barrier average was about 7.80 and the As well average was about 9.05. The difference between those two averages was taken as an empirical excess As signal. The cation/Ga well average was about 8.46. The three nearby cation/Ga neighbors then produced an empirical cation-to-As crosstalk coefficient of about 0.049, or about 4.9 percent.

The coefficient was then locally applied. The code searched for nearby cation/Ga neighbors for each As column using adjacent rows and summed their fitted peak volumes. The local summed cation/Ga volume was multiplied by the empirical coefficient to calculate the estimated crosstalk volume to assign to that As column. The output table reported for each As column the original estimated peak volume, the summed nearby cation/Ga volume, and the estimated crosstalk volume to subtract. The estimated crosstalk volumes were converted back into Gaussian peak amplitudes, placed at the As-column positions, convolved with the same probe/source kernel, and subtracted from the original ROI. The diagnostic output shows three images side by side: the original ROI, the estimated crosstalk-intensity image, and the intensity-corrected ROI. The estimated crosstalk image is seen to be spatially localized at the As-column positions and is expected to follow the same column geometry as the fitted model. The corrected ROI is seen to preserve the lattice structure while partially removing the modeled crosstalk contribution.

This output is notable because it changes the analysis from being a purely first-pass fitted-volume comparison into a second-pass image correction. The corrected ROI then becomes the input to the repeated fitting workflow. The corrected peak-volume and Al-fraction plots are therefore not just cosmetic adjustments of the first-pass result; they are produced after the analysis has been rerun on the corrected intensity image.

Table 5.3. Empirical cation-to-As crosstalk correction summary

Quantity	Approximate value / definition	Meaning
As barrier average	about 7.80	Reference As-column fitted volume in AlGaAs barrier rows.
As well average	about 9.05	Reference As-column fitted volume in GaAs well rows.
Excess As signal	$V_{\text{As,well}} - V_{\text{As,barrier}}$	Treated as empirical signal leakage from nearby cation/Ga columns.
Cation/Ga well average	about 8.46	Normalization reference for cation-to-As leakage.
Neighbor count	3 nearby cation/Ga columns	Local correction uses the three nearest adjacent-row cation/Ga neighbors.
$k_{\text{cat_to_As}}$	about 0.049 (4.9%)	Dataset-specific crosstalk coefficient; not a universal microscope parameter.

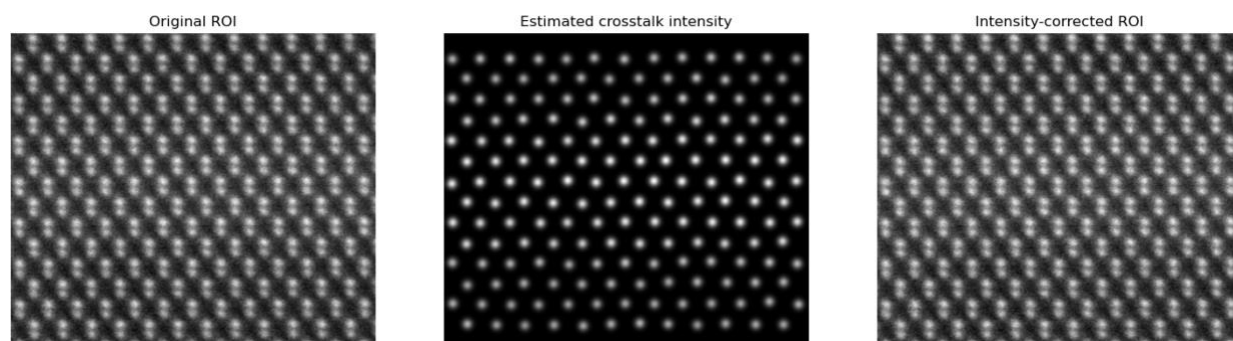


Figure 5.4. Crosstalk diagnostic image correction. Original ROI, estimated crosstalk contribution, and corrected ROI used for the empirical second-pass analysis.

5.9 Corrected volume and updated Al-fraction profile

After this corrected ROI was generated, the same fitting workflow was run on the ROI again. The corrected output was summarized in two ways. First, the row-averaged fitted peak volume was plotted as a function of row number for cation/Ga columns and As columns. Second, the individual fitted column volumes are plotted as a function of row position. The row-averaged plots show that the corrected workflow still produces a clear spatial trend through well and barrier. The cation/Ga fitted volumes remain on average higher near the GaAs well and lower toward the AlGaAs barrier regions, as expected from the stronger HAADF signal from Ga-rich cation columns compared with Al-containing barrier columns.

The corrected workflow then estimated Al fraction based on a local cation/As ratio. Instead of using only the cation fitted volume directly, each cation/Ga column was compared to nearby As columns. This local ratio was used because it partially normalizes shared local intensity changes and uses the As sublattice as a nearby reference. The GaAs well was taken as $x = 0$ and the AlGaAs barrier was taken as $x = 0.24$. Ratio values between these references were mapped into an estimated Al fraction.

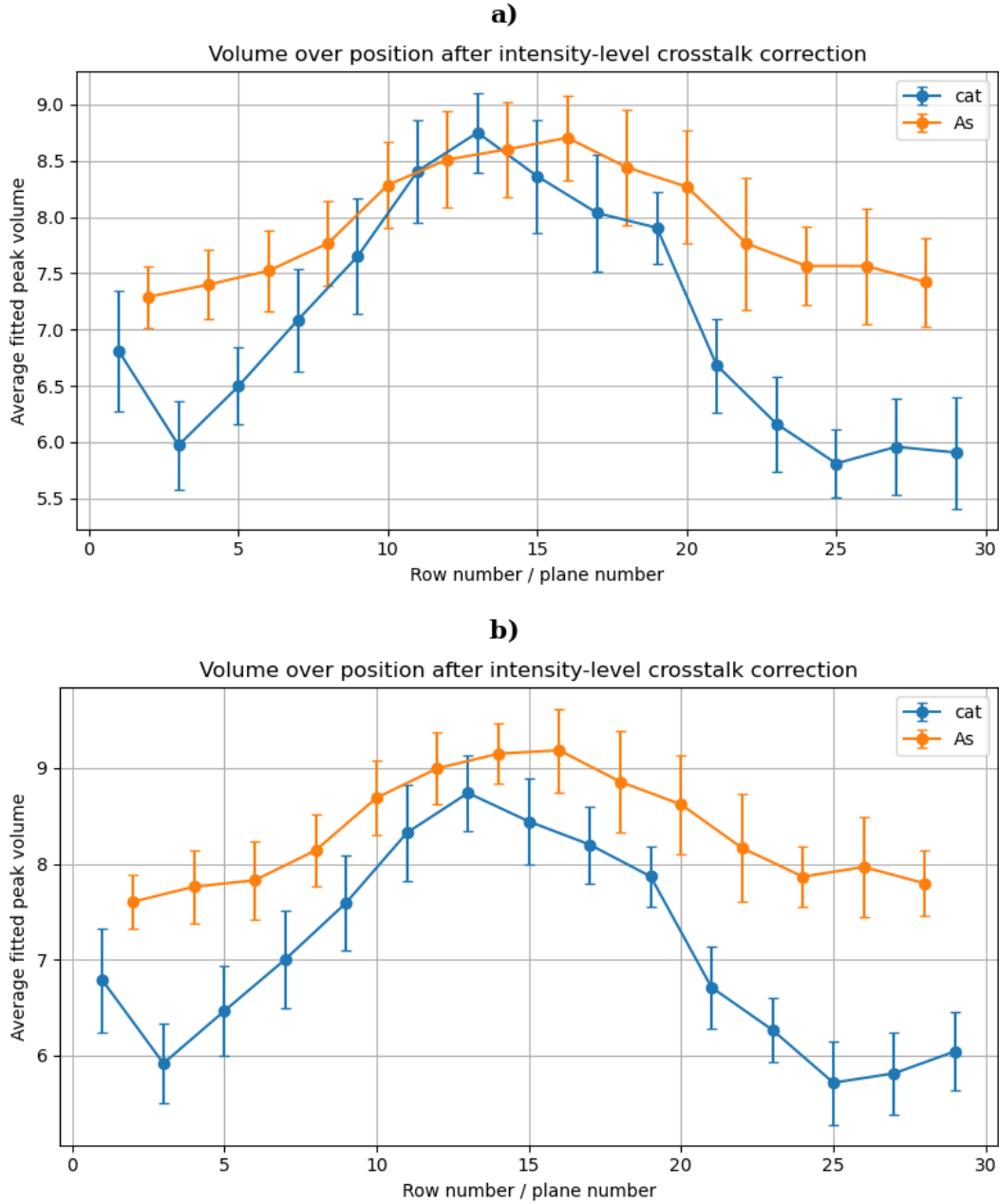


Figure 5.5. Corrected fitted-volume profiles by row after the empirical cation-to-As crosstalk correction and second fitting pass for a) Anscombe and b) PCA.

The resulting updated Al-fraction plot shows the resulting spatial profile through the analyzed rows. The dashed lower reference line corresponds to the GaAs well with $x = 0$ and the dashed upper reference line corresponds to the AlGaAs barrier with $x = 0.24$. The plotted points show the estimated row-level Al fraction after applying the crosstalk correction. Several rows near the well remain closer to the GaAs reference while rows closer to the barrier side

approach the expected AlGaAs reference. The error bars show that row-to-row and column-to-column variation remains substantial. Therefore the corrected profile should be interpreted as a relative trend and not an atom-by-atom composition map.

The main point of this corrected analysis is not that every atomic row can be assigned a unique exact composition. Rather, the result shows that the pipeline can produce a physically motivated second-pass correction, rerun the image fitting, and generate a corrected relative Al-fraction profile. This strengthens the thesis because it directly addresses one of the main limitations of raw fitted-volume interpretation: neighboring-column signal can bias the measured intensity of a target column. However, the corrected Al-fraction profile still requires cautious interpretation, as it depends on the empirical coefficient, the selected reference rows, the assumption of similar local thickness, and the accuracy of the probe/source kernel.

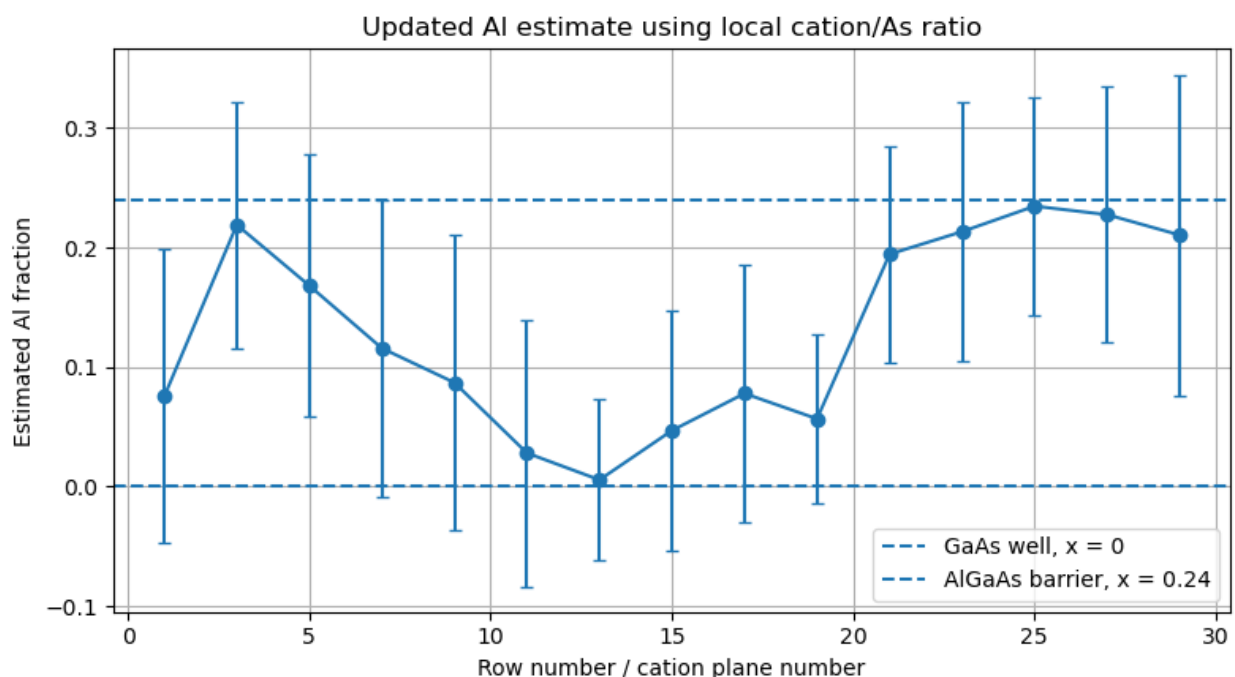


Figure 5.6. Updated relative Al-fraction profile based on the local cation/As ratio after correction for Anscombe. Dashed reference lines indicate the GaAs well and AlGaAs barrier endpoints.

5.10 Code output interpretation

The calculation outputs demonstrate that the thesis workflow can run from a denoised HAADF-STEM image all the way to interface localization, column detection, first-pass fitted peak

volumes, statistical comparison to reference regions, empirical crosstalk correction, second-pass fitting, and updated relative Al-fraction estimation. The full chained sequence is the primary technical contribution of the thesis. A crucial note on the interpretation of code output: The output should be interpreted in stages and not as a single final number.

The first-pass results establish a baseline. These results demonstrate that fitted peak volumes can be extracted from atomic columns and statistically compared between well, barrier, and interface regions. The tolerance intervals demonstrate that uncertainty is significant, especially where the reference distributions overlap. This result is itself scientifically useful because it discourages overconfident classification. A column whose fitted volume lies inside overlapping reference intervals should be reported as ambiguous rather than forced into a single chemical category.

The crosstalk-corrected results then address a specific limitation of the first pass. The finite probe and neighboring-column overlap can contribute intensity to nearby atomic-column positions, and so the corrected workflow estimates an empirical cation-to-As crosstalk term and subtracts its image-level contribution before rerunning the fitting process. The corrected ROI and updated Al-fraction profile show that this correction can be incorporated into the same modeling framework without having to abandon the original peak-volume logic.

The final interpretation is therefore cautious but stronger than the first-pass result alone. The workflow supports a relative composition trend across the GaAs/AlGaAs interface and yields a corrected profile that is more physically motivated than raw brightness or uncorrected peak volume alone. At the same time, the thesis does not claim that the corrected Al-fraction values are absolute composition measurements. The values remain preliminary relative estimates until they are validated against independent chemical measurements or more complete image simulations.

6. DISCUSSION

6.1 Logical progression of the research

The work unfolded in a manner that is in some sense scientifically logical. The initial stage gave the measured image physical meaning: electron scattering, diffraction, detector geometry, and HAADF Z-contrast. This was a prerequisite for discussion because composition cannot be discussed from a STEM image if the contrast mechanism is not understood. The next stage treated the image as numerical data and asked what effect noise has on the intensity distribution. This led to Fourier and PSD analysis, and provided a quantitative basis for comparison of denoising methods.

The third stage brought in Anscombe variance stabilization. Since STEM detector intensity has a relation to electron counts, Poisson-like noise should be expected. Anscombe processing was therefore a physically motivated preprocessing choice, rather than an arbitrary filter. The fourth stage took the work from image-level denoising to column-level modeling. This is a required stage since composition is not a property of the whole image, it must be inferred from the individual columns or planes of the interface.

The final stage used the known well and barrier regions as internal standards. This is the correct approach for the data in hand, because absolute HAADF quantification is limited by thickness and detector geometry. The net result is a thesis that progresses from electron optics to image processing, from image processing to model fitting, and from model fitting to careful relative composition estimation.

6.2 Scientific meaning of the Anscombe result

The Anscombe result should be interpreted as an improvement in preprocessing for this analysis pipeline. A denoising method should not be judged solely on its subjective visual smoothness but whether it preserves the information needed for the next stage. In this case, the next stage is atomic-column fitting. The PSD comparison shows that the Anscombe-based workflow reduces high-frequency noise and preserves the structural peaks. This is an indication that the Anscombe transformation is more compatible than PCA alone with the subsequent column-fitting workflow.

The result is not intended as an all-encompassing proof, as PCA performance is sensitive to the chosen components, the data representation, the noise level, and the image content. Anscombe performance is sensitive to whether the Poisson-like noise is a good approximation and whether the inverse transform introduces a bias. The thesis therefore uses carefully-worded language: For the available dataset and the radial PSD/profile metric used in this study, Anscombe-based processing performed better than PCA alone.

6.3 Scientific meaning of the composition output

The composition output indicates that fitted peak volumes can separate well and barrier references on average. The well reference has a larger mean fitted volume than the barrier references in the selected family. This is consistent with the expected HAADF contrast difference between GaAs and AlGaAs regions. Unknown interface columns can be placed on this reference scale.

The wide and overlapping tolerance intervals show that the present output should be interpreted as preliminary relative composition analysis rather than definitive column-by-column species identification. This is scientifically valuable because it reveals the uncertainty structure of the result. A less careful analysis might assign every column a number without indicating ambiguity. The present workflow instead shows when the data do not support a confident classification.

6.4 Scientific meaning and limits of the crosstalk correction

The crosstalk-correction workflow advances the scientific logic of the thesis because it more directly models the fact that atomic columns are not measured independently. If an ideal composition analysis were performed, the fitted intensity assigned to one column would come only from that column. In a real HAADF-STEM image, the finite probe shape, the convolution process, local blur, and neighboring columns all contribute to the measured intensity. This is particularly important near interfaces because both the composition and the neighboring columns may change rapidly and be in different chemical environments.

The empirical correction used As as a diagnostic reference sublattice. Since the main chemical change at a GaAs/AlGaAs interface is on the cation sublattice, an apparent As intensity

change between the well and barrier references may be caused by measurement contamination rather than an As-composition change. Estimating the excess As intensity and relating it to nearby cation/Ga fitted volumes, the correction workflow attempts to subtract off a portion of the neighboring-column contribution from the image before the second fitting pass.

This has two benefits. First, the coefficient is internally consistent because it is estimated from the same image, ROI, and fitted-volume quantities used in the rest of the thesis. Second, the correction remains compatible with the model-based framework because it is translated back into an image-level contribution using the same Gaussian peak-volume relationship and the same probe/source kernel. The correction therefore does not subvert the logic of the original fitting model.

On the other hand, the correction has important limitations. The crosstalk coefficient is an empirical and dataset-specific quantity. It assumes that the As well-barrier excess is primarily caused by neighboring cation/Ga crosstalk and not by thickness variation, residual scan distortion, incorrect region assignment, or another imaging artifact. It also assumes that the three nearby cation/Ga neighbors are an adequate local description of the crosstalk source. These assumptions are reasonable for a thesis-level refinement. However, they are not sufficient to claim a universal correction model. For this reason, the corrected Al-fraction profile should be treated as an improved relative estimate. It is more physically careful than the uncorrected fitted-volume result because it explicitly attempts to remove some of the neighboring-column intensity transfer. It is not, by itself, an independently validated composition map. A final publication-level claim would need to be compared to an independent chemical measurement, like EELS or EDS, or to image simulations with known composition, thickness, and microscope parameters.

6.5 Applicability to other material systems

While the applications shown in this thesis are specifically for GaAs/AlGaAs, the overall methodology is not chemically limited to Ga, Al, and As. Anscombe variance stabilization, atomic-column detection, probe-convolved peak fitting, and peak volume extraction all rely on knowledge of the image-formation process and columnar-intensity magnitudes instead of performing calculations on some prescribed set of elements. Several related quantitative HAADF-STEM methods have been demonstrated on other material systems, including InGaN/GaN and SiGe/Si [1], [10], [11].

These works indicate that composition-sensitive atomic-column analysis can be generalized to other materials provided that the columns are resolved and reference data or simulations are available. Transfer of this methodology is not immediate however, and the projected column families, reference compositions, probe model, and fitted-intensity-to-composition relationship will need to be re-established for each material system. In particular, the linear mapping to Al fraction and local cation/As ratio presented here are tailored specifically to GaAs/AlGaAs. Another material system would require a chemically inert reference sublattice, an alternate internal standard, calibration from EDS or EELS, or comparison with multislice simulations. The empirical cation-to-As crosstalk coefficient would also need to be derived again based on the new lattice geometry, specimen thickness, imaging conditions, and region of interest.

Overall, the methods presented here should therefore be most directly applicable to substitutional crystalline alloys that exhibit resolved atomic columns and sufficient HAADF contrast. Moving beyond these limits reduces reliability and HAADF-STEM analysis should be supplemented by spectroscopy or a light-element sensitive imaging technique such as iDPC-STEM. [12] While the overall framework can be transferred, each material system will require material-specific calibration, validation, and uncertainty analysis.

6.6 Shortcomings and risks

The first shortcoming is thickness. HAADF intensity is a function of both composition and sample thickness. Fitted intensity changes cannot be unambiguously interpreted as composition changes in the absence of an independent thickness map or a validated constant-thickness assumption. The relative-composition approach mitigates this problem by removing a common thickness variation but does not remove it entirely. The second shortcoming is probe calibration. The microscope parameters required for a completely calibrated probe model were not entirely available. The assumed aberration/source parameters are physically motivated but unverified, leaving the kernel not fully validated. This means the fitted volumes are not fully validated, and by extension the relative composition estimates.

The third shortcoming is classification overlap. The well and barrier tolerance intervals overlap substantially. Many unknown interface columns are therefore statistically ambiguous. The thesis reports this as a result, rather than producing an artificially certain classification.

The fourth shortcoming is validation. The present thesis does not include any independent EELS or EDS validation of the final estimated interface composition. This would be required in order to claim absolute values for the interface composition or a definitive physical mechanism of interface mixing.

The fifth shortcoming is authorship/workflow inheritance. The earlier HAADF workflow, image data, and some prior PCA-related analysis inherited the authorship and project context of an existing work. The present thesis uses them in that context, but only for background and comparison; it claims authorship only for the work actually performed by the author.

6.7 Recommended validation and completion steps

Before the final submission the numerical results should be compared to the exact final notebook output. The figure captions should be adapted to final saved figures rather than intermediate rendered pages. The committee should verify that relative Al-fraction values are to be kept as preliminary estimates or are to be moved to an appendix. If a final composition map is generated, the map should report uncertainty or classification ambiguity instead of just a single color scale.

The final thesis should also have a clear appendix containing the full code, but the main chapters should describe formulas, assumptions, inputs, outputs, and interpretation rather than display screenshots of code. This maintains a professional look and makes the scientific logic easier to follow.

7. CONCLUSIONS AND FUTURE WORK

7.1 Conclusions

This thesis presented an honest and reproducible workflow for HAADF-STEM analysis of GaAs/AlGaAs semiconductor interfaces. The work started from the physical understanding of electron scattering and HAADF Z-contrast, continued to numerical preprocessing, and ended at model-based atomic-column decomposition. The logical progression is important as composition cannot be responsibly estimated from raw brightness alone.

The preprocessing comparison demonstrated that Anscombe-based processing removed the high-frequency noise tail while leaving the structural peaks in the radial PSD/profile. This justified Anscombe preprocessing for this available dataset and the following column-fitting pipeline. The claim is deliberately limited to the dataset and metric studied here.

The author-written decomposition code modeled the HAADF image as a superposition of atomic-column peaks convolved with a probe kernel. Least-squares fitting resulted in column-specific peak volumes which were then compared across known well, known barrier, and unknown interface regions. The final output selected one common column family and produced reference means, tolerance intervals, classification plots, and relative Al-fraction estimates.

The thesis was augmented with an empirical crosstalk-correction workflow. It used the first-pass fitted volumes to estimate excess As-column intensity. Then, the local cation-to-As crosstalk coefficient was calculated. An estimated crosstalk-intensity image was constructed, subtracted from the ROI, and the fitting pipeline was rerun. The corrected volume-over-position plots and updated Al-fraction profile give a more physically cautious relative-composition output. The correction does not turn this method into an absolute composition measurement but does improve the interpretation by removing one source of neighboring-column intensity transfer.

The results demonstrate that the workflow can produce interpretable column-level outputs, but also that uncertainty is significant. Overlapping tolerance intervals result in some interface columns which cannot be uniquely classified. Therefore the thesis provides a robust computational framework and preliminary relative-composition analysis rather than an overconfident absolute composition measurement.

7.2 Future work

The future work should first benchmark the HAADF-derived composition estimates against an independent measurement such as EELS or EDS. Such benchmarking would indicate whether or not the estimated relative Al-fraction truly captures the physical interface composition.

Future work should more rigorously test the empirical crosstalk correction as well. The coefficient used in this case was determined from the present image. It should be tested for stability across multiple ROIs, multiple interfaces, and multiple images. A more complete study should compare the corrected and uncorrected Al-fraction profiles with independent chemical measurements and test the dependence of the corrected result on the assumed number of neighboring cation/Ga columns. If simulations with known GaAs/AlGaAs composition are available, the same correction workflow could be applied to simulated images to measure the ability of the method to recover the known input composition.

Second, thickness correction should be performed if an EELS thickness map or some other thickness estimate is available. This would better connect the fitted peak volume to composition.

Third, the probe kernel should be improved using verified microscope aberration and source parameters. A better kernel would limit model mismatch and thus the ability to reliably fit peak volumes. Fourth, the workflow should be used to analyze multiple images and multiple interface regions to test the reproducibility. A single dataset can show that the analysis is feasible, but a more extensive dataset is necessary for general conclusions.

Finally, uncertainty quantification should be expanded. Instead of reporting only the fitted values, future iterations of the pipeline should propagate uncertainty from denoising, column detection, fitting, reference selection and tolerance-interval classification. This would make the method more useful for publication-quality quantitative STEM.

A further future direction would be to calibrate the relative-composition workflow against multislice HAADF-STEM image simulations. In this study, GaAs/AlGaAs interface structures with a known Al/Ga distribution, thickness, and microscope parameters would be simulated under imaging conditions similar to the experiment. The same Anscombe preprocessing, column detection, peak-volume fitting, crosstalk correction, and relative Al-fraction extraction pipeline could then be run on the simulated images. As the simulated composition is known ahead of time, this comparison would indicate how well the workflow can recover the input interface profile and would allow one to separate errors from thickness, probe mismatch, column overlap, and noise.

Multislice simulation should thus be considered a future calibration and validation step, and not as evidence already presented in this thesis.

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APPENDIX A. CODE AND DATA BOUNDARY STATEMENT

Raw HAADF-STEM image data presented in this thesis work were provided as input data to the computational work. No authorship claim is made on previously collected microscopy data or workflows preexisting to the present thesis contribution. The only contribution made in this work is the Anscombe-based preprocessing usage and interpretation as well as model-based atomic-column decomposition workflow development for relative composition analysis of GaAs/AlGaAs interfaces.

The complete computational notebook and extended workflow outputs are not reproduced in full in this thesis because full code listings and long diagnostic tables would distract from the scientific logic of the analysis. Instead, the thesis reports the governing assumptions, input metadata, processing steps, fitted quantities, summary tables, selected outputs, and limitations needed to evaluate the work.

A copy of the broader project material is available at this link (subject to advisor approval and data-sharing restrictions on restricted materials).

APPENDIX B. SELECTED COMPUTATIONAL OUTPUTS

The main chapters contain the figures and compact tables necessary to read the scientific conclusions. Full notebook output, intermediate plots, long diagnostic tables, and code-cell detail are better stored and viewed as supporting computational material instead of as main-chapter text.

Repository Link:

<https://github.com/Behnamipurdue/Thesis-Project>

This appendix contains selected computational outputs and the source-repository location for the supporting workflow. The repository is included for reproducibility and context, while the body of the thesis remains focused on method, assumptions, results, interpretation, and limitations. Full code screenshots and long row-by-row tables are not included in the main text because they make the thesis difficult to read and are more appropriately placed in a repository or supplementary file.